1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In bladder cancer, high tumor-infiltrating lymphocyte (TIL) density is associated with improved cisplatin response only when the tumor circRNA and lncRNA landscape does not promote immune evasion. Specifically, elevated hsa_circ_0003221 in cancer stem cells drives stemness and immune-exclusion, attenuating TIL-mediated cytotoxicity and contributing to cisplatin resistance, whereas expression of lncRNA MEG3 supports tumor suppressive and immune-activating states that preserve chemosensitivity. Therefore, the predictive value of TIL density for cisplatin response is contingent on (i) low-to-moderate hsa_circ_0003221 expression and (ii) adequate MEG3 expression; combined assessment of these three factors (TIL density, hsa_circ_0003221, MEG3) provides superior prediction of therapy_response than TIL density alone. Mechanistically, hsa_circ_0003221 promotes CSC programs and immune evasion, potentially by miRNA sponging and modulation of pathways overlapping MEG3's tumor-suppressive/p53-related signaling; MEG3 loss removes a barrier to proliferation and dampens immune signaling. EIF4A3 and other RNA-processing factors may regulate circRNA abundance and thus modulate this axis. Testable predictions include: (a) in patient samples, the strongest cisplatin responders are TIL-high with low hsa_circ_0003221 and high MEG3; (b) in vitro, knockdown of hsa_circ_0003221 or overexpression of MEG3 increases cisplatin sensitivity in CSC-enriched bladder cancer models and enhances TIL/immune effector activity in co-culture; (c) in vivo, tumors with high circ_0003221 and low MEG3 show poor cisplatin response despite TIL presence, which can be rescued by targeting circRNA biogenesis or by MEG3 restoration or immune modulation.).

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	hsa_circ_0003221, lncRNA MEG3
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

hsa_circ_0003221: hsa_circ_0003221 induced/selected in cisplatin-treated bladder cancer stem cells → promotes proliferation/migration) because it directly links therapy exposure to an RNA-driven resistance/stemness program, and CORE PATH 2 (lncRNA MEG3 dysregulated via inflammation/miR-34a axis) because MEG3 provides a countervailing tumor-suppressive/immune-modulatory mechanism; together they form a plausible mechanistic toggle between chemosensitivity and resistance.

lncRNA MEG3: lncRNA MEG3 dysregulated via inflammation/miR-34a axis) because MEG3 provides a countervailing tumor-suppressive/immune-modulatory mechanism; together they form a plausible mechanistic toggle between chemosensitivity and resistance.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	hsa_circ_0003596	Supporting modifier from knowledge graph
Secondary	EIF4A3	Supporting modifier from knowledge graph
Secondary	circZFAND6	Supporting modifier from knowledge graph
Secondary	hsa-miR-424-5p	Supporting modifier from knowledge graph
Secondary	PRR15	Supporting modifier from knowledge graph
Secondary	HCP5	Supporting modifier from knowledge graph
Secondary	miR-34a-3p	Supporting modifier from knowledge graph
Secondary	seven-PAR-lncRNA	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- hsa_circ_0003596: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EIF4A3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.